

University of Minho

Master's Degree in [Computer Engineering / Artificial Intelligence / Data Engineering and Science]

Deep Learning

1st Year, 2nd Semester

Academic Year 2025/2026

Group Practical Assignment - Module 2

Instructor: Victor Alves

April 2026

English translation of the original Portuguese assignment brief.

Topic

Automatic Classification of ERCP Fluoroscopic Images via Deep Learning - Design and optimization of a computer vision solution for identifying biliary and pancreatic pathologies.

Work Context

Endoscopic Retrograde Cholangiopancreatography (ERCP) is a complex interventional procedure that uses fluoroscopy (real-time X-rays). Interpreting these images is challenging and crucial to the success of the procedure. The MIQR-CC dataset (Multimodal Image Quality and Risk in ERCP and Colonoscopy) provides thousands of curated images annotated by specialists, enabling the development of clinical decision support systems (CADx).

The objective of this work is to develop models capable of automatically classifying ERCP images into critical diagnostic categories:

- Biliary_Leaks (bile leaks)
- Lithiasis (biliary lithiasis, stones)
- Stricture (strictures, narrowing of the ducts)
- Normal (normal findings)

Learning Objectives

This practical assignment is intended to raise awareness and motivate students to solve concrete problems using the techniques covered in the course unit, applied to medical imaging. Specifically, students are expected to:

- Master the workflow for medical images (DICOM/PNG) and metadata.
- Apply, compare, and tune deep learning architectures (e.g., CNNs, Vision Transformers) for multi-class classification.
- Handle imbalanced datasets, which are common in medicine, through data augmentation, weighted loss functions, and asymmetric loss functions.
- Implement robust evaluation metrics (F1-score, confusion matrices, AUC-ROC).
- Ensure model reproducibility and interpretability (e.g., Grad-CAM).

Assignment Brief

Students are asked to develop a computational solution based on Deep Learning for the automatic classification of ERCP images, using as a basis the article "Curated endoscopic retrograde cholangiopancreatography images dataset (MIQR-CC)" and the corresponding dataset.

Article DOI: <https://doi.org/10.6084/m9.figshare.31079236>

Dataset download: <https://figshare.com/ndownloader/files/61063177>

The tasks include:

- Exploratory analysis: Study of class distribution and preparation of the dataset (train/validation/test split).
- Solution development: Use the solutions and results reported in the article and available in the GitHub repository as the baseline:
<https://github.com/monicacmartins/MIQR-CC-Dataset>.
- Optimization: Exploration of advanced techniques such as Transfer Learning, fine-tuning of SOTA models, or hybrid architectures.
- Evaluation: Rigorous comparison of results against the baseline presented in the article (macro F1-score of 0.738).
- Interpretability (mandatory): Generation of heatmaps to identify the image regions that influenced the model classification.

The AI system must be able to process input images and provide:

- Multi-class prediction: Probabilities for the 4 main classes;
- Robustness to noise: Handle contrast variations and artefacts common in X-rays;
- Visual explanation: Visualization of the detected critical areas (e.g., location of the stricture or stone).

Methodology

Students should base their development on the official repository and the provided data:

- Data access: The dataset is available via Figshare (<https://doi.org/10.6084/m9.figshare.31079236>). Students should use the processed versions of the images.
- Baseline review: Consult the code available on GitHub (<https://github.com/monicacmartins/MIQR-CC-Dataset>) to understand the preprocessing pipeline.

Suggested Workflow

- Preprocessing (e.g., resizing, normalization, CLAHE - Contrast Limited Adaptive Histogram Equalization).
- Implementation of strategies to address imbalance (e.g., class weights or oversampling).
- Training of candidate models.
- Validation and evaluation using the provided hold-out split (training, validation, and test).

Submission and Assessment

The results must be presented in a technical report (maximum 10 pages) written using the Springer conference paper template, containing:

- Image processing methodology (preprocessing).

- Detailed model architecture and hyperparameters.
- Comparative analysis (metric tables and confusion matrices), emphasizing the macro F1-score obtained on the test cases.
- Examples of interpretability visualization (Grad-CAM).

The report, together with a ZIP file containing the GitHub repository used, execution instructions (README), and requirements (requirements.txt), accompanied by examples and instructions that allow all steps and obtained results to be reproduced, must be submitted by one group member on the University of Minho e-learning platform under "Content/Module 2/Group Project - M2/Group Work Submission". Submissions must be completed by the end of the day on 31 May 2026.

During the week of 25 May to 30 May, during the class period, presentation and discussion sessions for the developed work will take place. The work groups must submit their presentations in advance on the University of Minho e-learning platform under "Content/Module 2/Group Project - M2/Presentation Submission". Each group will have 15 minutes to deliver the presentation, using whatever means it considers most appropriate.

Pedagogical Note

This work follows the challenge-based learning philosophy. Students are encouraged to:

- Try to exceed the baseline of the original article (F1-score > 0.738).
- Reflect on the clinical implications of false negatives in serious pathologies (e.g., strictures).
- Use experiment tracking tools (e.g., WandB or MLFlow) to document training progress.

Observation

The MIQR-CC dataset is particularly challenging due to the visual similarity between certain pathologies under fluoroscopy. The use of Attention Maps is suggested to verify whether the model is focused on the bile ducts or on external artefacts.

Peer Assessment

Each group must carry out a collective analysis of the contribution and effort that each member dedicated to the development of the work. Based on this analysis, the members who worked above average, at the average, and below average must be identified. For this assessment component, 1 point is assigned to each student, reflecting their individual contribution to the development of this assessment tool.

Therefore, one of the group members must send an email, with the other members in CC, to valves@di.uminho.pt. The subject line must be "AP[course] - TG - Avaliação Por Pares", where "course" must be replaced by the corresponding course (MEI, MIA, or MECD).

In the body of the email, for each group member, the delta must be indicated (the value to be added to the grade for this component). Note that deltas may be negative, zero, or positive and that, within each group, the sum of the deltas must always be equal to 0.00.

Example 1 (corresponds to equal effort by all):

PG1234 João DELTA = 0
 PG5678 António DELTA = 0
 PG9123 Maria DELTA = 0
 PG4567 Rita DELTA = 0

Example 2 (António receives 1 additional point, Rita keeps her grade, João and Maria each lose 0.5 points):

PG1234 João DELTA = -0.5
PG5678 António DELTA = 1
PG9123 Maria DELTA = -0.5
PG4567 Rita DELTA = 0

Code of Conduct

The participants in this academic work declare that they acted with integrity and confirm that they did not resort to plagiarism or to any form of improper use or falsification of information or results at any stage leading to its preparation. They further declare that they know and respected the Code of Ethical Conduct of the University of Minho.